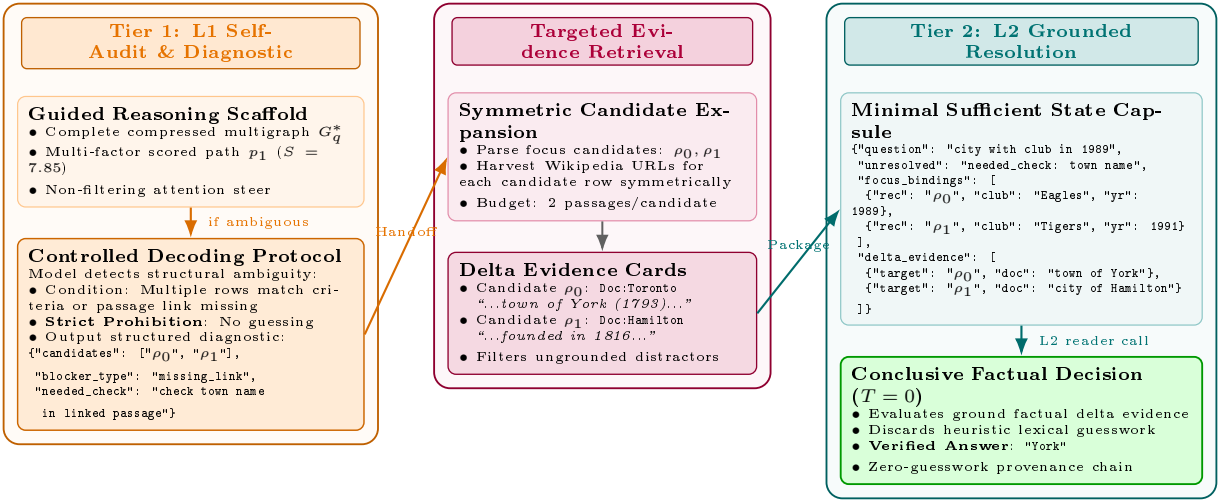

Tier 1: L1 Self-Audit & Diagnostic
Guided Reasoning Scaffold
• Complete compressed multigraph G_q^*
• Multi-factor scored path p_1 ($S = 7.85$)
• Non-filtering attention steer
if ambiguous
Controlled Decoding Protocol
Model detects structural ambiguity:
• Condition: Multiple rows match criteria or passage link missing
• Strict Prohibition: No guessing
• Output structured diagnostic:
{"candidates": ["ρ_0", "ρ_1"],
"blocker_type": "missing_link",
"needed_check": "check town name
in linked passage"}
Handoff
Targeted Evidence Retrieval
Symmetric Candidate Expansion
• Parse focus candidates: ρ_0, ρ_1
• Harvest Wikipedia URLs for each candidate row symmetrically
• Budget: 2 passages/candidate
Delta Evidence Cards
• Candidate ρ_0: Doc:Toronto
"...town of York (1793)..."
• Candidate ρ_1: Doc:Hamilton
"...founded in 1816..."
• Filters ungrounded distractors
Package
Tier 2: L2 Grounded Resolution
Minimal Sufficient State Capsule
{"question": "city with club in 1989",
"unresolved": "needed_check: town name",
"focus_bindings": [
{"rec": "ρ_0", "club": "Eagles", "yr": 1989},
{"rec": "ρ_1", "club": "Tigers", "yr": 1991}
],
"delta_evidence": [
{"target": "ρ_0", "doc": "town of York"},
{"target": "ρ_1", "doc": "city of Hamilton"}
]}
L2 reader call
Conclusive Factual Decision ($T = 0$)
• Evaluates ground factual delta evidence
• Discards heuristic lexical guesswork
• Verified Answer: "York"
• Zero-guesswork provenance chain